Assignment No. 2

Instructions : You have to submit me this assignment in hardcopy by Next week with explanation of every Algorithm.

Course: Concepts of Operating Systems (BSCS-502)
Topic: CPU Scheduling Algorithms
Total Marks: 10

This assignment is designed to help you understand and compare the working of different **CPU Scheduling Algorithms** used by operating systems to allocate CPU time to processes. Consider the following set of processes:

Process	Arrival Time (AT)	Burst Time (BT)	Priority
P1	0	8	2
P2	1	4	1
P3	2	9	4
P4	3	5	3
P5	4	2	5

Using the above process information, solve the scheduling problem using the following CPU Scheduling Algorithms: **First Come First Serve (FCFS), Shortest Job First (SJF – Non-Preemptive), Shortest Remaining Time First (SRTF), Priority Scheduling (Non-Preemptive), Priority Scheduling (Preemptive), and Round Robin (RR)** with a **time quantum of 3 milliseconds (ms)**. For each algorithm, first determine the order in which the processes are executed according to its scheduling rules. Then draw the corresponding **Gantt Chart** to illustrate the execution sequence of the processes on the CPU.

After drawing the Gantt Chart, calculate the **Completion Time (CT)** for each process, which is the time at which the process finishes execution. Next, calculate the **Turnaround Time (TAT)** using the formula **TAT = CT − AT**, where **AT** is the Arrival Time. Then calculate the **Waiting Time (WT)** using the formula **WT = TAT − BT**, where **BT** is the Burst Time. Also determine the **Response Time (RT)** using the formula **RT = First CPU Start Time − AT**, which represents the time from a process's arrival until it first gets CPU service.

Finally, compute the **Average Waiting Time** and **Average Turnaround Time** for each scheduling algorithm and compare the results. Based on your calculations, identify which algorithm

provides the best overall performance for this set of processes and justify your answer with appropriate reasons. Your comparison should consider factors such as average waiting time, average turnaround time, CPU responsiveness, and fairness among processes.

Instructions:

1. Solve all six scheduling algorithms using the given process table.
2. Draw a neat and correctly labeled Gantt Chart for each algorithm.
3. Show all calculations for Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), and Response Time (RT).
4. Calculate the Average Waiting Time and Average Turnaround Time for each scheduling algorithm.
5. Present your work in a clear and organized manner, showing all intermediate steps.
6. At the end of the assignment, write a brief comparison of all six algorithms and conclude which scheduling algorithm performs best for the given data, supporting your conclusion with your calculated results.